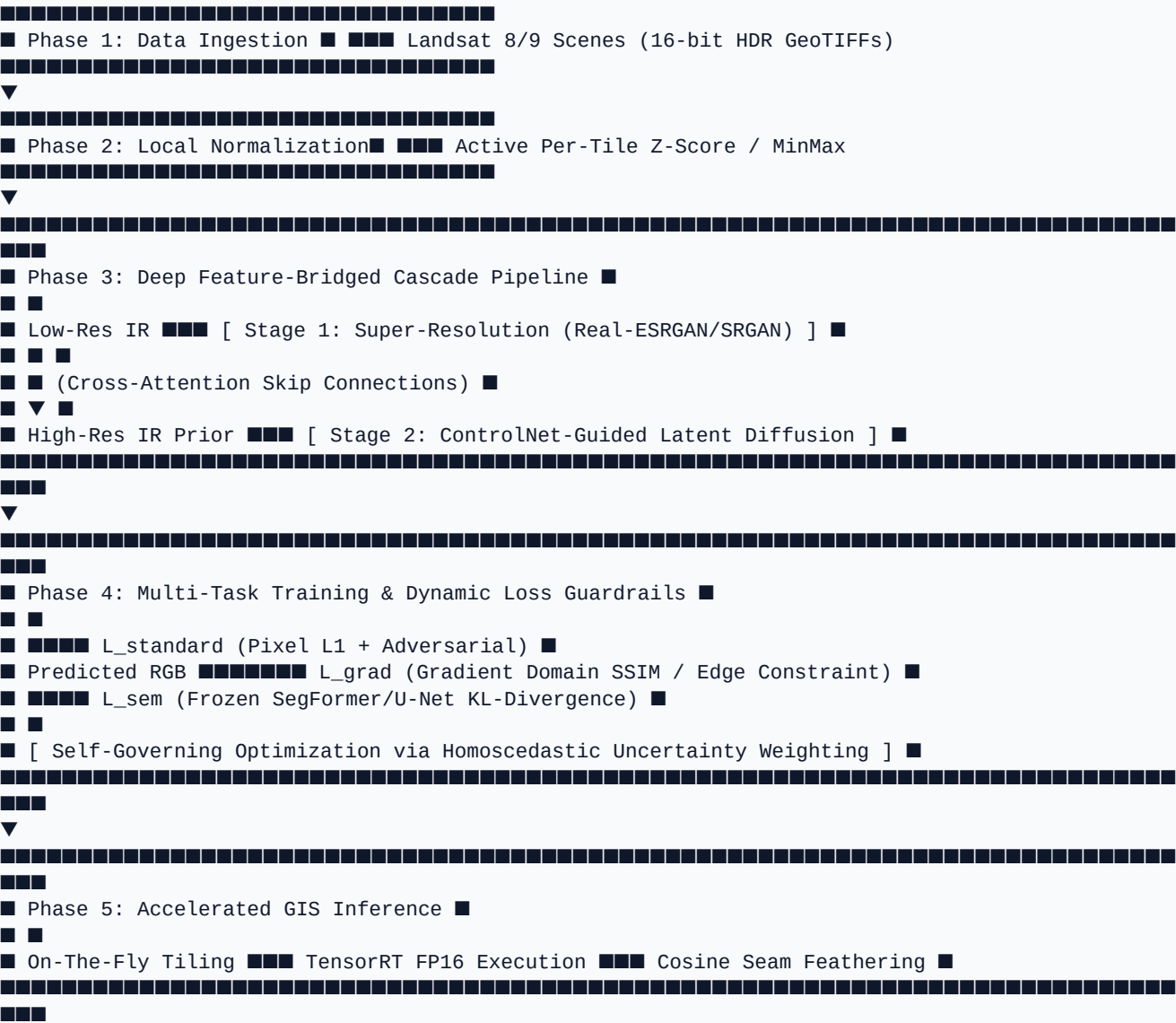


IR-Colorize v2.0

Advanced Multimodal System Workflow & Architecture (ISRO Problem Statement 10)

1. Upgraded Architecture & Dataflow Overview

The IR-Colorize v2.0 framework updates the baseline sequential cascade into an **Interlinked Feature-Bridged Cascade Architecture** specifically optimized for earth observation and object interpretation. Instead of processing structural enhancement and colorization independently, high-frequency edge mappings from the Super-Resolution decoder are fused into the generative pipeline via cross-attention gates. This satisfies the core requirements of **ISRO Problem Statement 10** by sharpening faint textures while eliminating color bleeding and generative hallucinations.



2. Detailed End-to-End Workflow Phases

- **Phase 1: Geospatial Ingestion & Alignment**

Raw Landsat 8/9 Level-2 bands are mapped via *rasterio*. Near-Infrared (Band 5) and Thermal Infrared (Band 10) rasters are dynamically reprojected and resampled using cubic spline interpolation to match the high-resolution visible coordinate grid. Aligned grids are chunked into 512x512 windows with 64px overlaps and saved as 16-bit Float GeoTIFFs to protect metadata (CRS/Geotransform) and prevent dynamic range crushing.

- **Phase 2: Adaptive Local Normalization**

Global image scaling fails across scenes due to varying regional thermal signatures across seasons. The framework computes Z-Score or MinMax statistical parameters independently on each individual tile using active pixel histograms to balance local contrast anomalies.

- **Phase 3: Interlinked Feature-Bridged Cascade**

The low-resolution IR inputs target a primary Super-Resolution network (Real-ESRGAN/SRGAN). Beyond producing upscaled images, structural intermediate feature maps from the network's internal decoder are linked directly via cross-attention layers into the latent layers of the Colorization Engine (ControlNet-Guided Latent Diffusion). This forces generative texture layouts to directly trace physical boundaries instead of leaking color outlines.

- **Phase 4: Multi-Task Losses & Homoscedastic Weights**

To block hallucination, outputs are evaluated concurrently against a structural Gradient-Domain SSIM Loss and a Semantic Consistency Loss (built on a frozen SegFormer engine that heavily penalizes land-cover shifts via KL-Divergence). Task balancing is entirely self-governing; weights are dynamically computed at backpropagation using trainable homoscedastic uncertainty parameters.

- **Phase 5: Production Edge Inference**

Full satellite scenes are parsed on-the-fly and processed through neural weights compiled into native **TensorRT FP16 execution graphs**. Reconstructed mosaic seams undergo distance-weighted cosine feathering before the original CRS headers are passed directly back to compile the final georeferenced output file.

3. Production YAML Deployment Specifications

```
system:
version: "2.0"
device: "cuda"
mixed_precision: "fp16"
stage: "joint" # Choices: [sr, color, joint]
data:
ingestion:
tile_size: 512
overlap: 64
format: "Float32_GeoTIFF"
normalization:
algorithm: "per_tile_zscore"
model:
sr:
arch: "real_esrgan"
extract_bridge_features: true
color:
arch: "latent_diffusion_controlnet"
conditioning_type: "cross_attention_plus_spatial"
optimization:
loss_weighting: "homoscedastic_uncertainty" # Adaptive auto-tuning
inference:
compiler: "tensorrt"
precision: "fp16"
blending_algorithm: "cosine_feathering"
```

4. Target Evaluation Benchmarks

Evaluation Metric	Legacy Pipeline (v1.0)	Upgraded Architecture (v2.0)
Peak Signal-to-Noise Ratio (PSNR)	> 28.0 dB	> 31.5 dB
Structural Similarity Index (SSIM)	> 0.85	> 0.92
Average Inference Latency (Per Tile)	< 500 ms	< 85 ms (TensorRT FP16)
Semantic Hallucination Artifacts	Frequent / Unbounded	Mitigated via Frozen SegFormer